

Quantum Photonic Computing for AI vs Classical Photonic AI Accelerators (2024–2025)

Executive Summary

Recent work positions quantum photonic computing as a longer-term, higher-risk but potentially higher-impact complement to classical photonic AI accelerators based on silicon photonics and related integrated platforms. Classical photonic accelerators and photonic interconnects are already at prototype or early-product stage for AI workloads, whereas photonic quantum processors are in the NISQ / early-fault-tolerant regime with commercial systems mainly targeting narrow sampling, optimisation, and generative-AI tasks.^{[1][2][3][4][5][6][7]}

Realistic roadmaps from PsiQuantum, Quandela, ORCA Computing, and Xanadu imply that broadly useful, error-corrected photonic quantum computers for general AI workloads are unlikely before the early-to-mid 2030s, even under optimistic assumptions. By contrast, silicon-photonics-based AI accelerators and optical interconnects are expected to scale into mainstream data-centre deployment across the second half of the 2020s and early 2030s.^{[8][9][10][11][12][7][13][14]}

1. Recent Technical Literature: Quantum Photonics for Machine Learning (2024–2025)

1.1 Algorithmic and systems papers

ProxiML: ML classifiers on photonic quantum hardware (2024). An ACM paper introduces ProxiML, a quantum machine-learning classifier implemented on Xanadu's X8 photonic quantum computer, achieving over 90% accuracy on four-class classification tasks and showing competitive performance versus other quantum platforms. The work emphasises noise-aware circuit design tailored to the loss and error characteristics of photonic hardware, demonstrating that practical ML workflows can already be run on small-scale photonic QPUs.^[^2]

Photonic probabilistic machine learning using quantum vacuum noise (2024). A Nature Communications paper demonstrates a photonic probabilistic computer based on a bistable optical parametric oscillator acting as a photonic probabilistic neuron, using quantum vacuum noise as a true randomness source. The authors implement probabilistic inference and MNIST digit generation using a time-multiplexed architecture and propose a fully optical system with projected sampling rates around 1 Gbit/s and energy consumption near 5 fJ/MAC, pointing to ultra-fast stochastic ML based on quantum optical noise.^[^15]

A linear photonic swap-test circuit for quantum kernel estimation (2024). An arXiv preprint reports an integrated photonic circuit implementing the swap-test algorithm to estimate inner

products of quantum states, a key ingredient for quantum support-vector machines and kernel methods. The device performs swap-test-based kernel estimation on thousands of input pairs with mean error around 0.04, demonstrating that photonic circuits can implement non-trivial QML primitives with good accuracy.[¹⁶]

Error-mitigated photonic quantum circuit Born machine (2025). A Physical Review A Letter presents a quantum circuit Born machine (QCBM) designed for photonic hardware and shows that “recycling mitigation” can substantially improve training under realistic photon-loss noise. Experiments on a quantum photonic integrated processor illustrate that error-mitigation techniques tailored to loss can materially enhance the performance of photonic generative models.[¹⁷]

1.2 Platform and architecture papers

A versatile single-photon-based quantum computing platform (Quandela, 2024). A Nature article associated with Quandela describes a quantum processor based on high-efficiency quantum-dot single-photon sources, universal interferometers, and superconducting nanowire detectors, implemented as a cloud-accessible photonic platform. The work demonstrates multi-photon operations (up to six photons at a few hertz) and characterises key performance metrics, positioning the platform as a candidate for near-term photonic quantum algorithms, including QML.[¹⁸][19]

A manufacturable platform for photonic quantum computing (PsiQuantum-linked, 2024 preprint; 2025 Nature). A widely cited PsiQuantum-authored paper describes monolithically integrated silicon-photonics-based modules for qubit generation, manipulation, networking, and detection, achieving state-preparation-and-measurement fidelities near 99.98%, high-visibility Hong–Ou–Mandel interference, and two-qubit fusion gates with more than 99.2% fidelity. The authors argue that the demonstrated building blocks constitute a “feature-complete” photonic technology stack compatible with future fault-tolerant cluster-state architectures.[³][20]

Scaling and networking a modular photonic quantum computer (2025). Another Nature paper discusses a modular photonic architecture in which photonic-chip modules are networked via fibre to form a universal, fault-tolerant machine once component performance reaches target thresholds. The article notes that current demonstrations such as the Aurora system confirm scalability of photonic approaches but still fall short of full error-corrected operation.[⁸]

Linear optics to scalable photonic quantum computing (2025). A review-style arXiv paper surveys the evolution from linear-optics demonstrations to current large-photon-number experiments and discusses machine-learning-assisted design of photonic circuits, time- and frequency-multiplexing, and other scalability techniques. The authors highlight boson sampling with more than 100 photons as evidence of growing computational power, while stressing that losses and component imperfections remain central obstacles to practical large-scale PQC.[²¹]

1.3 Photonic quantum-accelerated ML

Photonic quantum-accelerated machine learning with Quandela Ascella (2025). A recent arXiv preprint introduces a quantum optical reservoir computing (QORC) scheme implemented on Quandela's Ascella photonic QPU, showing that hybrid quantum-classical reservoirs can match or exceed small classical neural networks while requiring up to twenty-fold fewer training samples on tasks such as MNIST and medical image classification. The authors argue that high-dimensional boson-sampling-type dynamics can provide genuine acceleration by reducing data and epoch requirements, and they demonstrate close agreement between simulations and hardware experiments.[⁴]

Photonic quantum ML under noise (2026, outlook-relevant). A 2026 overview paper on photonic quantum machine learning analyses various QML algorithms adapted to photonic platforms and emphasises that advances such as hybrid superconducting-photonic chips with loss resilience near 99% are crucial for scaling. The work underscores that many photonic-QML experiments to date run on relatively small systems, with quantum advantage contingent on further loss mitigation, multiplexing, and better encodings.[²²]

Integrated-photonics QML and ONNs review (2024). An MDPI review article surveys quantum computation and optical neural networks implemented on integrated photonics, including photonic quantum processors and classical optical neural networks, and discusses implemented algorithms. It frames integrated photonics as a unifying hardware substrate spanning quantum and classical photonic computing for machine learning.[²³]

2. Recent Technical Literature: Classical Photonic AI Accelerators (2024–2025)

2.1 Silicon-photonics AI accelerator demonstrations

Integrated large-scale photonic accelerator with ultralow latency (Nature, 2025). A Nature paper reports a 64×64 photonic accelerator implemented on a 65-nm commercial silicon-photonics platform co-packaged with a 28-nm CMOS electronic integrated circuit. The chip integrates more than 16,000 photonic components and delivers matrix-multiply-accumulate (MAC) operations up to 1 GHz with an average numerical precision of 7.61 bits, demonstrating practical performance on optimisation tasks such as max-cut.[⁷]

Photonic integrated-circuit platform for AI acceleration (IEEE JSTQE, 2025). An IEEE Journal of Selected Topics in Quantum Electronics article describes a heterogeneous photonic platform that integrates III-V compound semiconductors with silicon to realise wafer-scale optical neural-network building blocks (lasers, amplifiers, modulators, detectors, non-volatile phase shifters) on a single chip. The authors argue that this platform can outperform state-of-the-art electronic accelerators in energy efficiency while being manufacturable for datacentre AI workloads.[¹³]

Photonics for sustainable computing (2024). An arXiv paper develops a carbon-footprint model for photonic accelerators, analysing ADEPT, a photonic deep-neural-network accelerator, and concluding that photonics can reduce both operational and embodied carbon relative to 28-nm CMOS, partly because fabrication has roughly four times lower

carbon cost per unit area. This study strengthens the case that classical photonic accelerators address not only performance but also sustainability constraints in large-scale AI.[¹]

2.2 Industry perspectives and market outlook

Photonics and AI: industry perspectives (OPN, 2024). An Optics & Photonics News feature interviews companies including Lightmatter and discusses how silicon photonics is being used both for optical interconnects and as an accelerator for neural-network computations. Lightmatter's strategy couples a photonic compute engine with a high-bandwidth photonic interconnect fabric, and the article notes strong investor interest and a mission to reduce the energy footprint of hyperscale AI.[²⁴]

Lightmatter Passage and silicon-photonics interposers (2024). A 2024 analysis in *The Next Platform* describes Lightmatter's Passage silicon-photonics interposer, designed to link compute and memory dies with high-bandwidth optical connections and claimed by the CEO to underpin "all of the GPUs in the world" for AI over the next few years. The piece positions Passage and similar photonic fabrics (e.g., Celestial AI's Photonic Mesh) as near-term solutions to AI's data-movement bottleneck in training very large models.[⁹]

Silicon photonic AI accelerator market reports (2025). Market studies project strong growth for silicon-photonics-based AI accelerators and photonic integrated circuits from 2025 to the mid-2030s, driven by AI workloads, quantum computing, and LiDAR, among other applications. A 2025 HPC and AI accelerator market report similarly highlights photonic accelerators as an emerging category in future datacentre architectures.[²⁵][²⁶][¹⁴][²⁷]

2.3 Related photonics-for-computing reviews

Photonics perspective on computing with physical substrates (2024). An Elsevier review article takes a broad view of photonics-based computing, including neuromorphic and analogue photonic systems, and discusses their potential to alleviate von-Neumann bottlenecks in AI workloads. It situates photonic AI accelerators alongside other unconventional computing substrates.[²⁸]

Photonics for neuromorphic computing (2024). An arXiv review on photonics for neuromorphic computing outlines how integrated photonics and optoelectronic devices can implement neural-network primitives and emphasises the synergy with deep-learning workloads. The review notes that the exponential growth in AI model size and data volumes is a key driver for photonic approaches.[²⁹]

3. Company Status and Roadmaps in Quantum Photonic Computing

3.1 PsiQuantum

PsiQuantum is pursuing a fault-tolerant photonic quantum computer based on silicon photonics, cluster states, and fusion-based quantum computing (FBQC), targeting on the order of one million physical qubits. The company's public statements and third-party analyses suggest an ambition to build commercially useful, error-corrected machines in the

late 2020s or early 2030s, with DARPA's US2QC/QBI programme setting an upper timeline of around 2033 for an “industrially useful” quantum computer.^{[10][11]}

PsiQuantum's 2024–2025 technical work includes the Omega chipset, a manufacturable photonic platform fabricated in GlobalFoundries' high-volume CMOS foundry, and a detailed FBQC architecture tolerant to photon-loss probabilities above 10% per fusion event. These advances address manufacturability and error-tolerance but still require large-scale system integration and sustained improvements in loss, source brightness, and detector efficiency.^{[20][30][31][3]}

3.2 Quandela

Quandela operates cloud-accessible photonic quantum processors such as Ascella, which uses a high-efficiency quantum-dot single-photon source, universal interferometer, and superconducting detectors to implement universal linear-optical circuits. The platform supports demonstrations of multi-photon logic gates and algorithms and is used as the hardware backend in recent photonic ML experiments, including quantum optical reservoir computing.^{[19][32][4][18]}

In 2024 Quandela published a 2024–2030 technology roadmap targeting first logical (error-free) qubits by 2025, around 50 logical qubits by 2028, and “industrialised” error-corrected quantum computers by 2030. The roadmap emphasises scaling quantum operations per second by roughly 25× (from 400 to 10,000 QOPS) and leveraging quantum networking to connect multiple photonic processors, with hybrid HPC–quantum deployments such as the Lucy system integrated into the Joliot-Curie supercomputer for optimisation, chemistry, and ML workloads.^{[12][33]}

3.3 ORCA Computing

ORCA Computing builds photonic quantum systems based largely on standard telecoms components combined with atomic memories, targeting near-term hybrid AI applications rather than fully error-corrected universal QC. Its PT-series systems, including PT-1 and PT-2, are deployed in government and industrial environments and integrate with NVIDIA GPUs via platforms such as CUDA-Q to run hybrid quantum–classical workloads.^{[5][34][35][6]}

ORCA focuses on generative AI and optimisation, with its CEO publicly targeting quantum advantage on specific AI problems within about 24 months, emphasising that its systems aim to accelerate existing generative models rather than replace classical hardware. Recent announcements include a PT-2 testbed for quantum-enhanced generative AI, deployments at the UK National Quantum Computing Centre, and blueprints for data-centre integration where photonic QPUs act as accelerators alongside Hopper-class GPUs.^{[34][35][^5]}

3.4 Xanadu

Xanadu develops continuous-variable photonic quantum computers and the PennyLane software stack for hybrid quantum–classical ML, with cloud-accessible hardware such as the X8 system used in ProxiML and other QML experiments. In 2025 the company unveiled Aurora, described in industry coverage as a modular universal photonic quantum computer

comprising multiple networked photonic processors operating at or near room temperature.
[³⁶][37][²]

Xanadu's strategy includes deep partnerships (e.g., with Applied Materials, Corning, and defence and aerospace organisations) to improve fabrication yields, optical interconnects, and size-weight-and-power metrics, positioning photonic quantum systems for eventual data-centre deployment. A 2026 joint research initiative with Lockheed Martin aims to explore photonic-hardware implementations of generative QML models that exploit Fourier-type operations believed to be hard for classical kernels, illustrating the company's focus on QML as a first major application domain.[³⁷][38][³⁹][40]

4. Classical Silicon Photonics for AI: Status and Roadmaps

4.1 Current deployment modes

Classical photonics for AI is currently commercialised mainly in two roles: high-bandwidth interconnect (co-packaged or chip-to-chip optics) and analogue/neuromorphic compute accelerators for matrix operations. Silicon photonics is already pervasive in datacentre optical transceivers and is extending into intra-package and intra-rack interconnect for GPU clusters, with companies such as Lightmatter and Celestial AI promoting photonic fabrics to overcome bandwidth and energy bottlenecks in large AI training clusters.[⁹][7][²⁴]

Prototypes of photonic AI accelerators have been demonstrated in CMOS-compatible foundry processes, with integrated optical MAC arrays reaching GHz-scale operation and multi-bit precision in 64×64 configurations. Research systems based on heterogeneous III-V/silicon integration suggest that full optical neural networks with on-chip lasers and non-volatile phase shifters are feasible at wafer scale, offering superior energy efficiency over electronic accelerators.[⁷][13]

4.2 Industry and market expectations

Industry commentary and market analyses anticipate significant adoption of silicon-photonics-based AI interconnect and accelerators from the mid-2020s onward, driven by the exponential growth of AI models and datacentre energy constraints. Lightmatter's leadership, for example, has publicly suggested that its Passage photonic fabric could underpin the majority of AI-class GPUs "over the next few years," signalling aggressive near-term deployment ambitions.[²⁶][29][²⁵][24][⁹]

Market reports for silicon photonic AI accelerators foresee multi-billion-dollar global market sizes by the early-to-mid 2030s, aligning with more general forecasts for HPC and AI accelerators where photonics is listed as an emerging category. Silicon-photonics advances in 400-Gb/s-per-wavelength links and AI-assisted equalisation also show that the ecosystem is solving the bandwidth and signal-integrity challenges required for large-scale deployment.
[¹⁴][41][²⁷][25][²⁶]

5. Comparative Technical Maturity (2024–2025)

5.1 Hardware scale, fidelity, and programmability

Photonic AI accelerators based on classical optics currently achieve tens of thousands of integrated components per chip, multi-GHz MAC rates, and effective precisions of 7–8 bits, using commercial silicon-photonics and CMOS technologies. Their operation is largely classical, with optical interference used for linear algebra and electronic control loops and ADCs providing non-linearity and programmability.^{[13][1][^7]}

Photonic quantum processors, in contrast, typically operate with tens of modes and a handful of indistinguishable photons, with system-level performance limited by source brightness, loss, and detector efficiency. Recent work has demonstrated very high single- and two-qubit operation fidelities ($\geq 99\%$) and manufacturable building blocks, but end-to-end error-corrected logical qubits have not yet been realised in photonic hardware.^{[42][21][3][18][20][8]}

5.2 AI workloads and demonstrated applications

Classical photonic accelerators have demonstrated acceleration for optimisation problems, vector-matrix multiplications, and certain ML inference tasks, with some platforms explicitly targeting transformer-like workloads and optical neural networks for inference and even training. However, integration into mainstream ML frameworks and toolchains remains nascent, and many demonstrations focus on relatively small models or stylised tasks.^{[43][29][^7]}

Photonic quantum systems have been used for QML experiments such as ProxiML classifiers, swap-test-based kernel estimation, QCBM generative models, and quantum optical reservoir computing, often matching classical baselines on moderate-size datasets rather than conclusively outperforming them. ORCA's PT-series and Quandela's Ascella are being integrated into hybrid workflows with NVIDIA GPUs and HPC systems to target generative AI, optimisation, and sampling tasks, but evidence for clear, scalable quantum advantage in real-world AI use cases is still emerging.^{[6][33][2][17][16][4][5][34]}

5.3 Tooling and ecosystem

For classical photonic AI, the ecosystem leverages mature CMOS design flows, silicon-photonics PDKs, and existing HPC infrastructure, with companies offering hardware that plugs into standard racks and leverages familiar software stacks. This lowers barriers to deployment, even though programming models for analogue photonic accelerators (e.g., calibration, mapping of weights to interferometer meshes) are still an active research area.^{[43][29][24][14][^9]}

In photonic quantum computing, software stacks such as PennyLane and Perceval, as well as vendor-specific SDKs, provide high-level interfaces for designing QML experiments, but compilation to fault-tolerant architectures and resource estimation for large-scale QML remain complex and unsettled. Hybrid GPU-QPU frameworks such as NVIDIA's CUDA-Q integration with ORCA platforms and HPC coupling of Quandela's Lucy system illustrate

early moves toward a fuller ecosystem, but the stack is far less mature than classical photonic AI toolchains.^{[44][33][36][42][5][34]}

6. Realistic Timeline: Quantum Photonic vs Classical Photonic AI

6.1 Time-horizon comparison

The table below synthesises indicative timelines based on technical results and company roadmaps. These are necessarily approximate and assume continued progress but no unforeseen breakthroughs.

Timeframe	Classical photonic AI (silicon photonics, ONNs, interconnect)	Quantum photonic computing for AI (PQCs, QML)
2024–2026	Pilot deployments of photonic interposers and optical fabrics in AI GPU clusters (e.g., Lightmatter Passage, Celestial AI), continued prototyping of on-chip photonic accelerators in foundry processes, and early commercial systems targeting specific inference/optimisation tasks. ^{[9][7][^24]}	Commercial NISQ-regime photonic QPUs (Ascella, PT-series, Xanadu X8) available via cloud/on-prem, running QML demos (ProxiML, swap-test kernels, QCBM, QORC) and integrated with GPUs and HPC for exploratory AI workloads. ^{[2][4][5][6][^33]}
2027–2030	Expected broader deployment of photonic interconnects and perhaps first production use of photonic AI accelerators for specific datacentre workloads, as indicated by market forecasts and ongoing IEEE/Nature-level prototypes. ^{[7][13][25][14]}	Vendor roadmaps target first logical qubits (Quandela 2025) and tens of logical qubits with early error-corrected processors by ~2030, with focus on niche AI/optimisation problems where sampling or reservoir-like dynamics offer advantage. ^{[12][8][10][4]}
2030–2035	Silicon-photonics AI accelerators and optical interconnects likely to be standard options in hyperscale datacentres, especially for bandwidth-bound training and inference, with multi-billion-dollar market size. ^{[25][26][14][27]}	Ambitious goals from PsiQuantum, Quandela, and others envision industrialised error-corrected photonic quantum computers with hundreds of logical qubits and potential to tackle non-toy AI workloads, contingent on sustained gains in loss, source brightness, and system integration; DARPA’s US2QC/QBI programme uses 2033 as a target horizon for an “industrially useful” machine. ^{[10][11][12][8]}
Beyond 2035	Further scaling of photonic accelerators and close integration with electronic AI silicon, possibly including fully optical or hybrid neuromorphic systems, as well as more sophisticated photonic co-processors for transformers and other architectures. ^{[28][29]}	If error-corrected photonic quantum systems mature, they could become specialised accelerators for sampling-heavy generative models, probabilistic inference, and certain kernel or linear-algebra operations where asymptotic quantum advantage can be realised, but this remains speculative and heavily dependent on progress this decade. ^{[15][22][4][45]}

6.2 Key factors shaping the timeline

Loss and component performance. Quantum photonic computing must contend with compounding photon loss across large circuits, making ultra-low-loss waveguides, high-brightness sources, and near-unity-efficiency detectors essential; recent work reaches impressive component metrics but has not yet closed the full system-level gap required for large-scale error correction. Classical photonic AI accelerators, by contrast, can tolerate

much higher loss because signals can be amplified and noise corrected at the electronic interface.^{[3][42][20][8]}

Manufacturability and integration. PsiQuantum's use of GlobalFoundries and the demonstration of manufacturable photonic-quantum chipsets show that CMOS-foundry integration is possible, but assembling millions of components into a coherent, error-corrected system with cryogenic detectors and complex control remains a formidable engineering challenge. Silicon-photonics AI platforms benefit from decades of investment in datacentre optics and can re-use infrastructure for packaging, testing, and deployment, which accelerates commercial timelines.^{[30][31][41][20][25][9]}

Algorithmic advantage and workload fit. Photonic QML systems are focusing on domains where sampling, generative modelling, or reservoir-like dynamics might yield genuine quantum advantage with relatively modest logical-qubit counts, but clear, scalable advantage over classical methods has yet to be established. Classical photonic accelerators, meanwhile, target linear algebra at scale and are naturally aligned with the dominant workloads in today's AI (transformers, diffusion models), offering a clearer path to near-term value.^{[2][15][22][4][43][7][^13]}

Ecosystem and tooling. The classical photonic AI stack is converging with existing HPC practices, while quantum photonic stacks still require specialised expertise and lack mature debugging, verification, and resource-estimation tools, which will slow large-scale deployment. Progress in high-level QML frameworks and hybrid GPU-QPU platforms will be necessary to make photonic quantum accelerators practically usable for AI practitioners.^{[44][36][5][14]}

7. Takeaways for Positioning Quantum Photonics as a Parallel Path

Quantum photonic computing is best viewed, for the next decade, as a specialised accelerator path running in parallel with rapidly maturing classical photonic AI hardware rather than as a direct near-term replacement. Classical silicon-photonics AI accelerators and interconnects are likely to dominate practical AI deployments through at least the early-to-mid 2030s, while photonic quantum systems explore niche QML and sampling applications and work toward error-corrected architectures.^{[4][5][12][2][9][7][^13]}

From a research and investment standpoint, this suggests allocating effort both to classical photonic accelerators—where deployment risk is lower and timelines shorter—and to quantum photonic platforms, particularly in areas like probabilistic ML, boson-sampling-based accelerators, and reservoir computing where credible paths to advantage may emerge with relatively modest logical-qubit resources.^{[45][15][16][4]}

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